

FlowSentinel AI

Autonomous Data Engineering Operations & Resilience Framework

⚡ Executive Summary

Traditional Data Engineering on-call is broken—engineers spend **over 40% of their time** firefighting stack traces, chasing late vendor feeds, and manually running backfills at 2:00 AM.

FlowSentinel AI transforms Data Ops on GCP by mirroring Google’s internal SRE stack (Composer, BigQuery, Monitoring, Vertex AI Gemini 2.0). **AI agents handle 90% of the heavy lifting** (RCA, feed monitoring, backfill calculation, code PRs), while human leaders retain 1-click governance.

🔥 Competitive Edge: FlowSentinel AI vs Standard Demos

Operational Metric	Standard Industry Demos	FlowSentinel AI Framework
Incident Response (MTTR)	2 to 6 hours (Manual debugging)	< 2 minutes (Instant AI RCA & fix)
Human Involvement	High execution fatigue	1-Click Governance (Approve via Dashboard)
Source Feed Downtime	Corrupts downstream tables	Proactive Sentinel (Auto-pauses DAGs)
Code-Change Backfill	Manual guesswork & quota risk	Blast-Radius Engine (Lineage & cost plan)
Ticketing Integration	Disconnected manual entry	Jira / GitHub Sync (Auto RCA & PR link)

🔧 Core Framework Pillars



1. Autonomous RCA & Triage

Gemini 2.0 intercepts pipeline alerts, parses stack traces & git history, drafts root-cause analysis, and logs enriched Jira tickets in seconds.



2. Source Feed Sentinel

Monitors GCS/SFTP feed arrival SLAs. Pauses downstream DAGs on delay, alerts vendors, and auto-resumes when files land.



3. Precision Backfill Engine

Parses DAG lineage on code diffs, maps affected partitions, calculates BigQuery slot costs, and prepares 1-click backfill plans.



4. Dual-Persona Command Deck

React/Vite app on Cloud Run + Looker Studio. Features real-time pipeline status (🟢/🟡/🔴), feed timers, and 1-click execution cards.



ROI & Business Impact

⚡ **85% MTTR Reduction:** Hours of debugging down to seconds.

🛡️ **70% On-Call Savings:** Eliminates manual off-hours intervention.

🛡️ **Zero Data Poisoning:** Sentinel prevents incomplete dashboard loads.

🚀 **30-Min Portability:** Terraform-ready for any client GCP tenant.

🎬 Live Demo Narrative (3:00 AM Pitch Script)

"An upstream vendor changes a CSV schema at 3:00 AM, breaking BigQuery. Instead of waking up an engineer:

- 1. FlowSentinel identifies schema drift and creates a structured Jira ticket with the exact line of fault.*
- 2. The Code Repair Agent drafts a GitHub PR with the updated schema mapping.*
- 3. The Backfill Engine calculates 7 days of historical partitions need re-running (\$1.20 cost).*
- 4. The Lead Engineer taps **[Approve PR & Backfill]** on their phone and goes back to sleep."*

GCP Stack: Cloud Composer 3, BigQuery, Dataform, Cloud Logging, Pub/Sub, Vertex AI Gemini **Ticketing:** Jira / GitHub
2.0 Issues